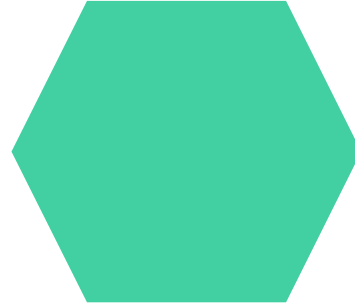
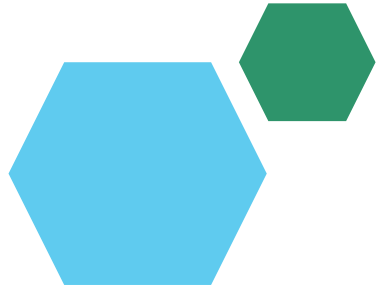


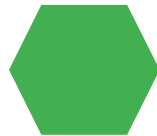


Project Title - seasonal-agriculture-performance-analysis

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College Name- G.Narayanamma Institute of Technology and Science]

AICTE STU ID: STU6a5fc4dc72bec1784661212



PROBLEM STATEMENT

- (Seasonal variations can affect agricultural performance. However, raw agricultural data may not clearly show how crop performance changes across different seasons and farming conditions.

This project analyzes the dataset to identify seasonal patterns, relationships, variations and economic outcomes.)



Project Description

For this project, I looked at how crop performance changes across the three farming seasons in India — Kharif, Rabi, and Zaid. The idea was to understand what actually drives the differences: things like rainfall, temperature, soil condition, the irrigation method used, and how much fertilizer or pesticide goes into each crop. I didn't stop at just yield and production either — I wanted to see the money side too, so I looked at revenue, cost, and profit for each season and crop.

Before doing any real analysis, I had to clean up the dataset since it had missing values — I filled those in using season-wise and crop-wise medians so the numbers stayed realistic instead of just using an overall average. Once the data was ready, I used Python for the whole analysis — Pandas and NumPy to work with the numbers, and Matplotlib and Seaborn to visualize the patterns. I also ran a correlation analysis to see which factors were most closely tied to yield, and finished with an ANOVA test to check whether the yield differences between seasons were actually statistically meaningful or just random variation.

WHO ARE THE END USERS?

- **Farmers** — can use this to get a clearer sense of which crop tends to do better in which season, before deciding what to plant
- **Agricultural planners and policymakers** — can use these patterns to guide decisions around resource allocation and planning
- **Agri-businesses and government agriculture departments** — can track how performance shifts season to season and plan accordingly
- **Researchers and students** — get a data-backed look at how seasonal conditions actually affect farming outcomes, instead of relying on assumptions

Technology Used

- Python
- Pandas & NumPy — data cleaning and analysis
- Matplotlib & Seaborn — visualizations (bar charts, scatter plots, heatmaps, box plots)
- SciPy — statistical testing (ANOVA)
- Jupyter Notebook / Google Colab — development environment
- Excel dataset (.xlsx) as the data source



RESULTS

9. Seasonal Analysis

```
season_analysis = df.groupby("Season")[[
    "Yield_Tonnes_Ha",
    "Production_Tonnes",
    "Revenue_INR",
    "Profit_INR"
]].mean()
```

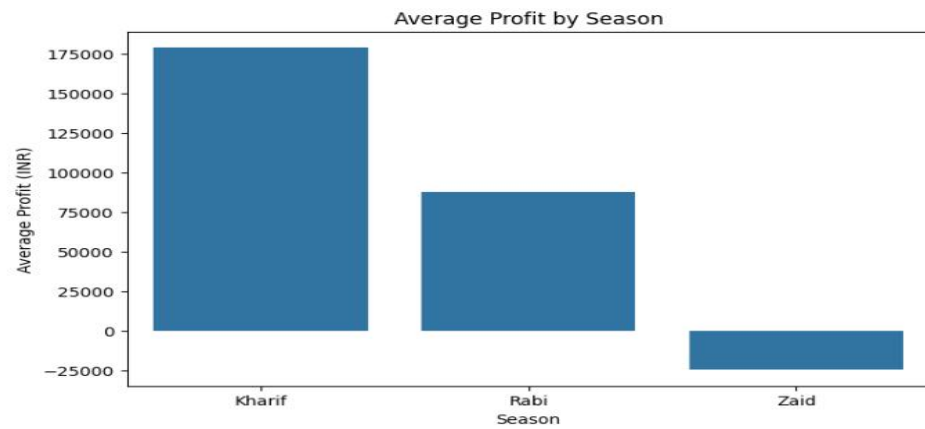
season_analysis

	Yield_Tonnes_Ha	Production_Tonnes	Revenue_INR	Profit_INR
Season				
Kharif	5.630616	46.311192	710719.055649	178914.647555
Rabi	5.094917	41.486712	601526.048556	87689.470191
Zaid	4.637340	38.886111	519171.904040	-24804.824916

```
plt.figure(figsize=(8,5))
```

```
sns.barplot(
    x=season_analysis.index,
    y=season_analysis["Profit_INR"]
)
```

```
plt.title("Average Profit by Season")
plt.xlabel("Season")
plt.ylabel("Average Profit (INR)")
plt.show()
```

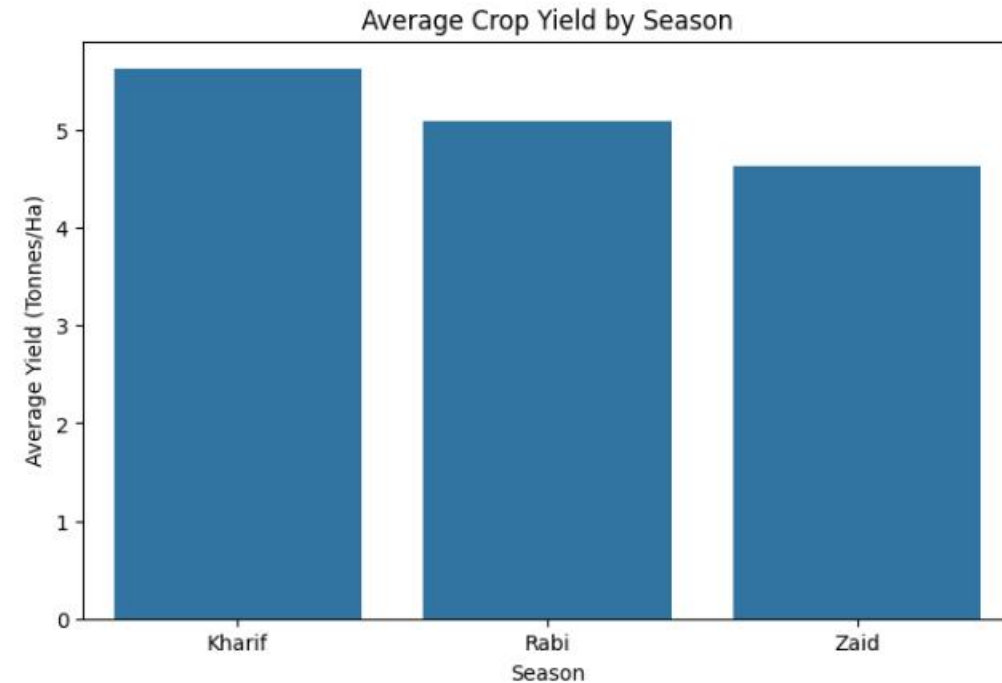


```
plt.figure(figsize=(8,5))
```

```
sns.barplot(
    x=season_analysis.index,
    y=season_analysis["Yield_Tonnes_Ha"]
)
```

```
plt.title("Average Crop Yield by Season")
plt.xlabel("Season")
plt.ylabel("Average Yield (Tonnes/Ha)")
plt.show()
```

...



9.1 Seasonal Production Comparison

```
season_production = df.groupby("Season")["Production_Tonnes"].mean()

print("Average production by season:")
print(season_production)
```

```
... Average production by season:
Season
Kharif    46.311192
Rabi      41.486712
Zaid      38.886111
Name: Production_Tonnes, dtype: float64
```

RESULT

10. Crop vs Season Analysis

```
crop_season = df.groupby(["Crop", "Season"])[ "Yield_Tonnes_Ha"].mean().unstack()
```

crop_season

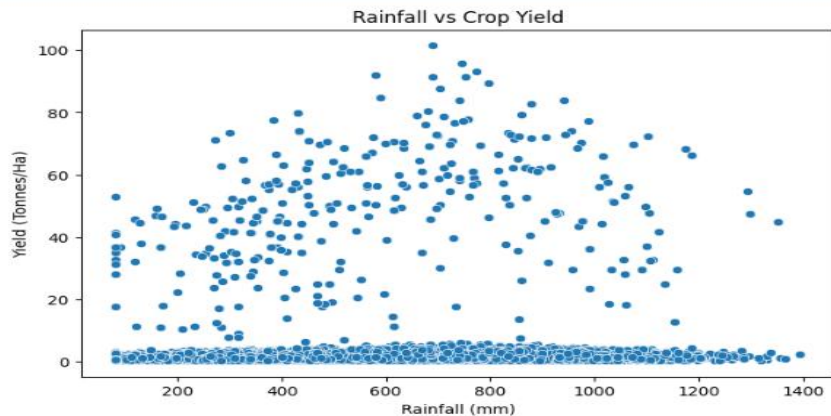
Season	Kharif	Rabi	Zaid
Crop			
Chilli	1.731676	1.462822	1.179062
Cotton	1.373140	1.192985	0.951848
Groundnut	1.483834	1.222994	1.037778
Maize	2.966410	2.617489	2.297262
Pulses	1.036063	0.872019	0.652258
Rice	2.709268	2.334526	1.900490
Sugarcane	53.459120	43.963953	38.423922
Wheat	2.255788	2.059515	1.748588

12. Rainfall vs Yield

```
plt.figure(figsize=(8, 5))

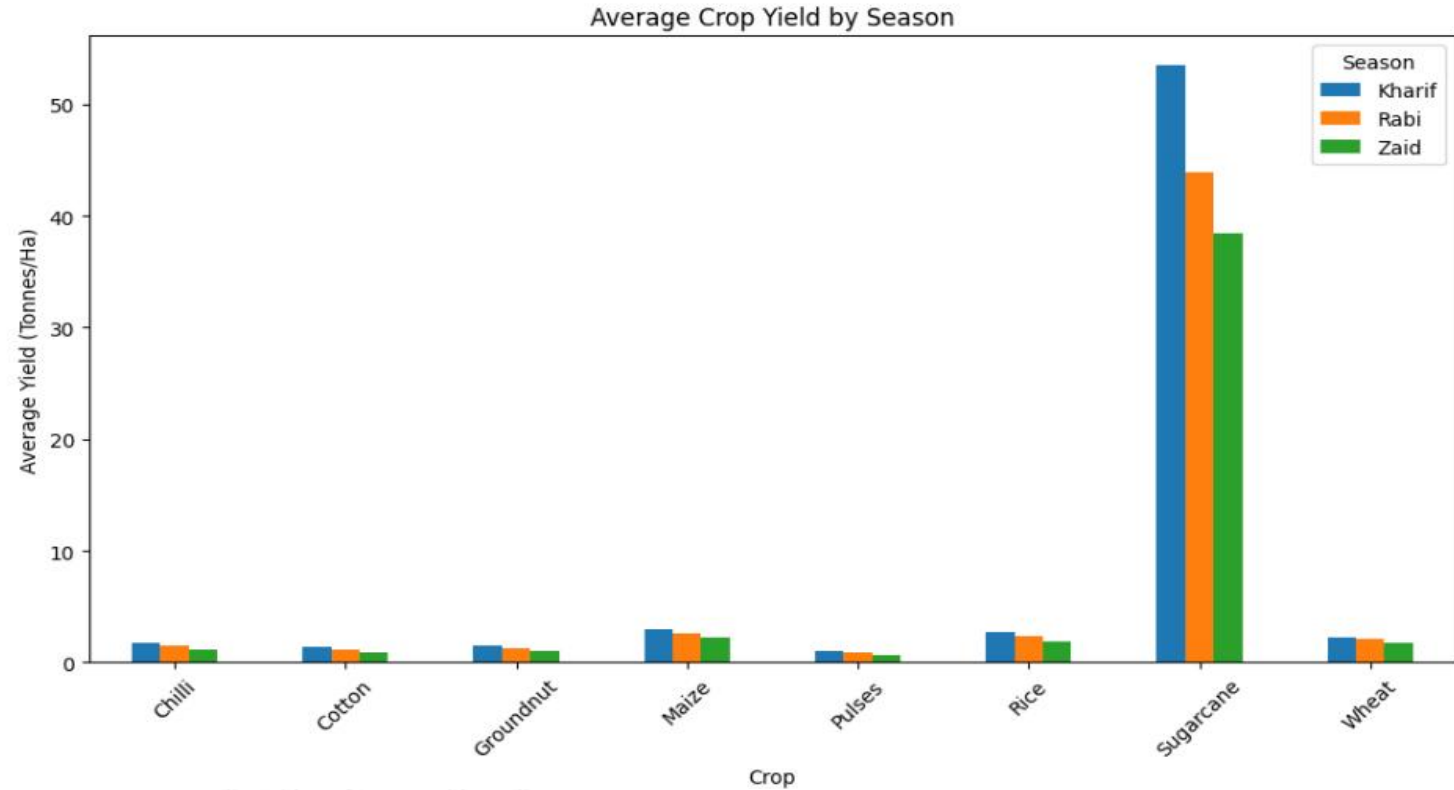
sns.scatterplot(
    data=df,
    x="Rainfall_mm",
    y="Yield_Tonnes_Ha"
)

plt.title("Rainfall vs Crop Yield")
plt.xlabel("Rainfall (mm)")
plt.ylabel("Yield (Tonnes/Ha)")
plt.show()
```



```
crop_season.plot(kind="bar", figsize=(12, 6))

plt.title("Average Crop Yield by Season")
plt.xlabel("Crop")
plt.ylabel("Average Yield (Tonnes/Ha)")
plt.xticks(rotation=45)
plt.legend(title="Season")
plt.show()
```



19. Statistical Analysis

```
from scipy.stats import f_oneway

kharif = df[df["Season"] == "Kharif"]["Yield_Tonnes_Ha"]
rabi = df[df["Season"] == "Rabi"]["Yield_Tonnes_Ha"]
zaid = df[df["Season"] == "Zaid"]["Yield_Tonnes_Ha"]

test_result = f_oneway(kharif, rabi, zaid)

print("ANOVA result:")
print("F-statistic:", test_result.statistic)
print("p-value:", test_result.pvalue)
```

```
ANOVA result:
F-statistic: 1.4579031041437143
p-value: 0.2328474889882346
```

RESULTS

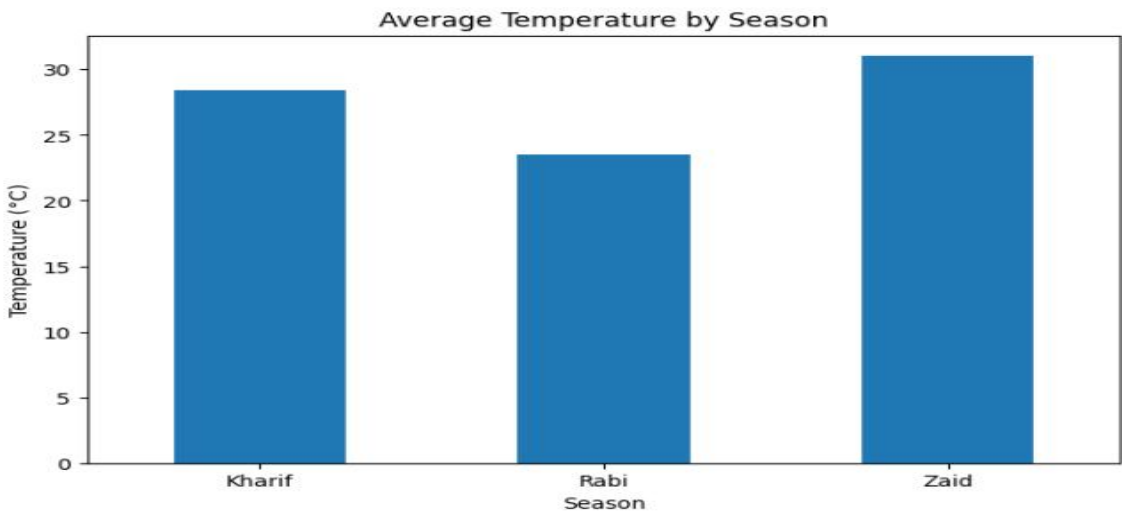
11. Environmental Analysis

```
environment = df.groupby("Season")[[
    "Rainfall_mm",
    "Avg_Temperature_C",
    "Humidity_pct",
    "Soil_Moisture_pct",
    "Sunlight_Hours_Day"
]].mean()
```

	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Soil_Moisture_pct	Sunlight_Hours_Day
Season					
Kharif	852.112676	28.454019	71.812591	31.198426	6.793367
Rabi	435.937062	23.489183	57.893178	24.050215	7.589797
Zaid	299.123232	31.042088	52.012121	19.169024	8.178788

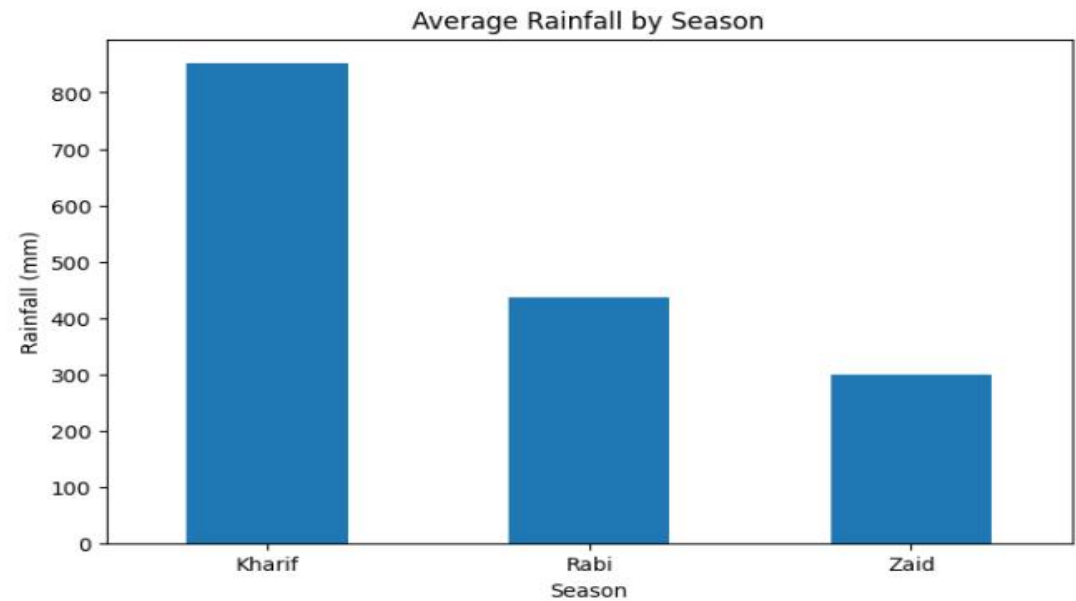
```
environment["Avg_Temperature_C"].plot(kind="bar", figsize=(8, 5))

plt.title("Average Temperature by Season")
plt.xlabel("Season")
plt.ylabel("Temperature (°C)")
plt.xticks(rotation=0)
plt.show()
```



```
environment["Rainfall_mm"].plot(kind="bar", figsize=(8, 5))

plt.title("Average Rainfall by Season")
plt.xlabel("Season")
plt.ylabel("Rainfall (mm)")
plt.xticks(rotation=0)
plt.show()
```

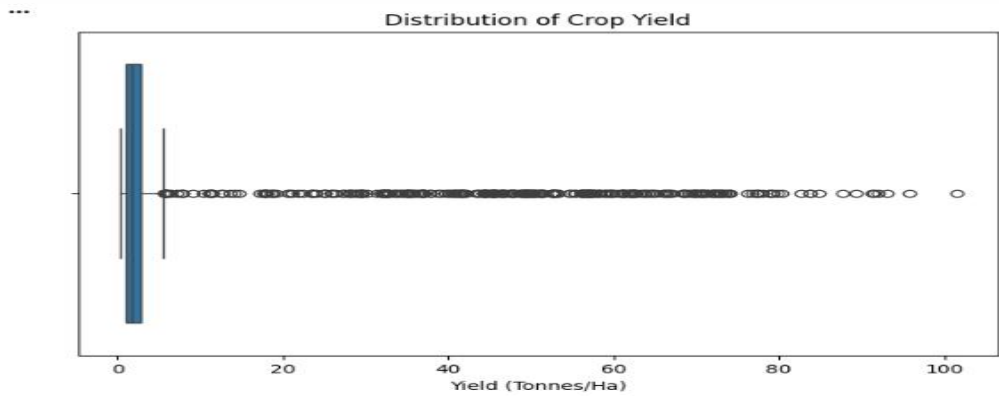


20. Unusual Patterns

```
plt.figure(figsize=(8, 5))

sns.boxplot(
    x=df["Yield_Tonnes_Ha"]
)

plt.title("Distribution of Crop Yield")
plt.xlabel("Yield (Tonnes/Ha)")
plt.show()
```



RESULTS

14. Irrigation & Water Usage

```
irrigation_count = df["Irrigation_Method"].value_counts()
```

```
print("Irrigation methods used:")  
print(irrigation_count)
```

```
Irrigation methods used:  
Irrigation_Method  
Flood      1310  
Rainfed    1041  
Drip        915  
Sprinkler   734  
Name: count, dtype: int64
```

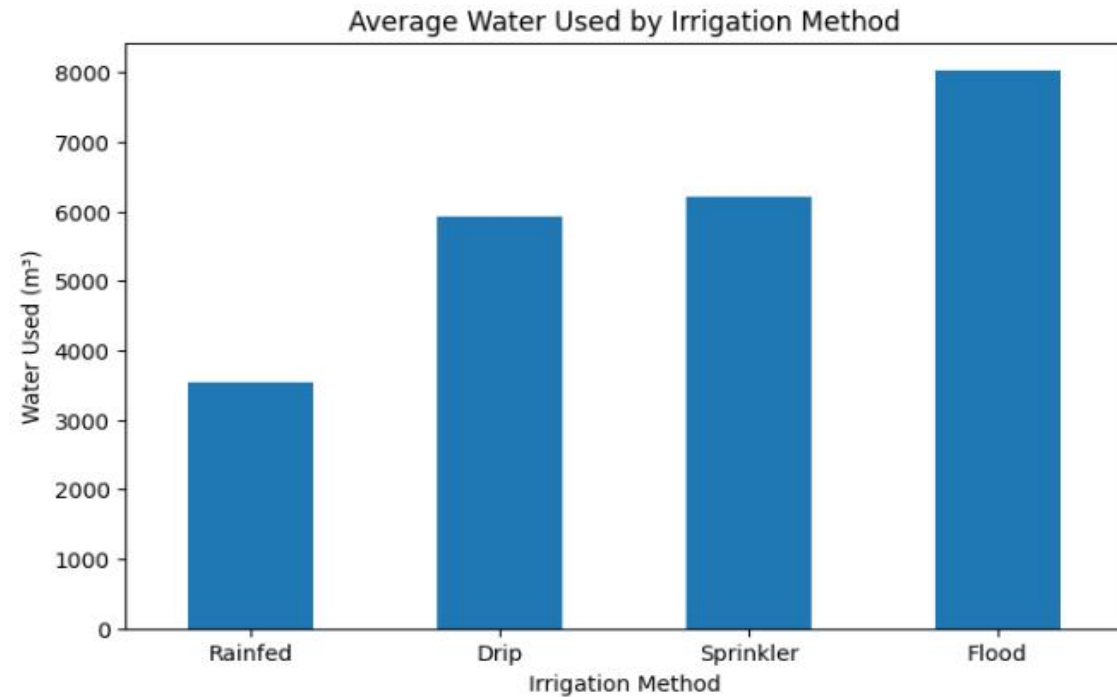
```
water_check = df.groupby("Irrigation_Method")["Water_Used_m3"].mean()
```

```
print("Average water used by irrigation method:")  
print(water_check)
```

```
Average water used by irrigation method:  
Irrigation_Method  
Drip      5918.747541  
Flood     8026.472519  
Rainfed   3549.554275  
Sprinkler 6208.258856  
Name: Water_Used_m3, dtype: float64
```

```
water_check.sort_values().plot(  
    kind="bar",  
    figsize=(8, 5)  
)
```

```
plt.title("Average Water Used by Irrigation Method")  
plt.xlabel("Irrigation Method")  
plt.ylabel("Water Used (m³)")  
plt.xticks(rotation=0)  
plt.show()
```



15. Fertilizer & Pesticide Analysis

```
fertilizer_season = df.groupby("Season")["Fertilizer_kg_ha"].mean()
```

```
print("Average fertilizer usage:")  
print(fertilizer_season)
```

```
Average fertilizer usage:  
Season  
Kharif    187.077965  
Rabi      185.407068  
Zaid      184.638047  
Name: Fertilizer_kg_ha, dtype: float64
```

16. Disease/Pest Risk Analysis

```
risk_season = df.groupby("Season")["Disease_Pest_Risk_pct"].mean()
```

```
print("Average disease and pest risk by season:")  
print(risk_season)
```

```
Average disease and pest risk by season:  
Season  
Kharif     54.467903  
Rabi       40.482114  
Zaid       38.216667  
Name: Disease_Pest_Risk_pct, dtype: float64
```

RESULTS

18. Correlation Analysis

```
numeric_data = df[[  
    "Rainfall_mm",  
    "Avg_Temperature_C",  
    "Humidity_pct",  
    "Soil_Moisture_pct",  
    "Fertilizer_kg_ha",  
    "Pesticide_Litre_ha",  
    "Yield_Tonnes_Ha",  
    "Production_Tonnes",  
    "Revenue_INR",  
    "Total_Cost_INR",  
    "Profit_INR",  
    "Water_Used_m3",  
    "Disease_Pest_Risk_pct"  
]]
```

```
correlation_table = numeric_data.corr()
```

```
correlation_table.round(2)
```

	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Soil_Moisture_pct	Fertilizer_kg_ha	Pesticide_Litre_ha	Yield_Tonnes_Ha	Production_Tonnes	Revenue_INR	Total_Cost_INR	Profit_INR	Water_Used_m3	Disease_Pest_Ris
Rainfall_mm	1.00	0.20	0.53	0.52	0.02	0.00	0.03	0.02	0.10	0.01	0.11	0.00	
Avg_Temperature_C	0.20	1.00	0.14	0.08	0.02	0.01	0.01	0.01	0.03	0.04	0.01	0.02	
Humidity_pct	0.53	0.14	1.00	0.45	0.02	-0.00	0.01	0.00	0.06	0.01	0.06	-0.00	
Soil_Moisture_pct	0.52	0.08	0.45	1.00	0.01	-0.02	0.01	0.01	0.08	0.00	0.09	0.00	
Fertilizer_kg_ha	0.02	0.02	0.02	0.01	1.00	-0.00	-0.00	0.00	0.01	0.15	-0.07	0.00	
Pesticide_Litre_ha	0.00	0.01	-0.00	-0.02	-0.00	1.00	-0.01	-0.01	-0.00	0.05	-0.03	0.00	
Yield_Tonnes_Ha	0.03	0.01	0.01	0.01	-0.00	-0.01	1.00	0.89	0.43	0.03	0.49	0.39	
Production_Tonnes	0.02	0.01	0.00	0.01	0.00	-0.01	0.89	1.00	0.56	0.19	0.55	0.52	
Revenue_INR	0.10	0.03	0.06	0.08	0.01	-0.00	0.43	0.56	1.00	0.52	0.89	0.43	
Total_Cost_INR	0.01	0.04	0.01	0.00	0.15	0.05	0.03	0.19	0.52	1.00	0.07	0.58	
Profit_INR	0.11	0.01	0.06	0.09	-0.07	-0.03	0.49	0.55	0.89	0.07	1.00	0.19	
Water_Used_m3	0.00	0.02	-0.00	0.00	0.00	0.00	0.39	0.52	0.43	0.58	0.19	1.00	
Disease_Pest_Risk_pct	0.62	0.25	0.55	0.38	0.02	-0.47	0.01	0.01	0.06	0.01	0.07	0.01	

Future scope

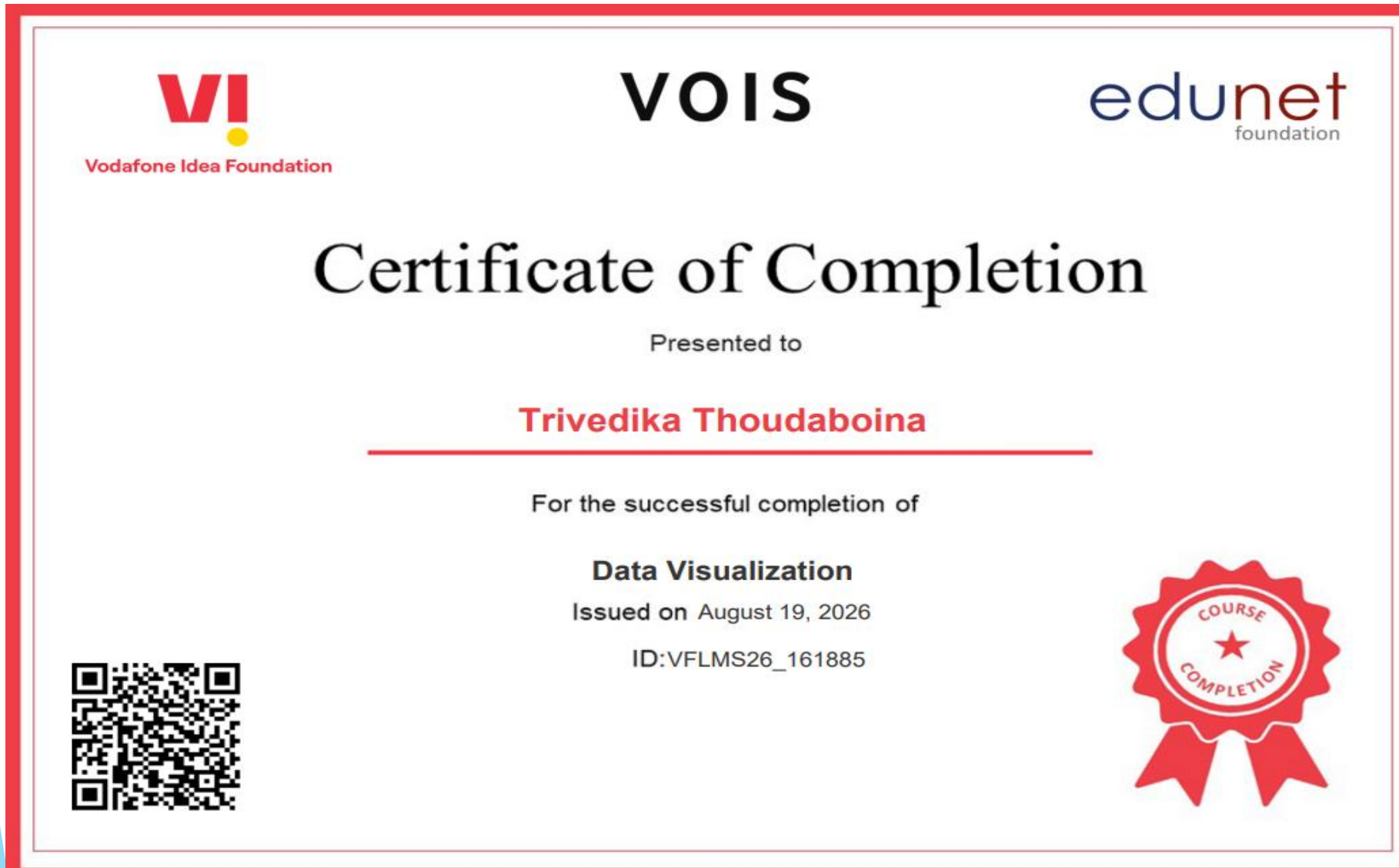
- Bringing in real-time weather and rainfall data, along with forecasts, would keep the analysis current and let it look ahead instead of only backward
- Building an interactive dashboard (something like Power BI) would let people explore the data themselves instead of just seeing fixed charts
- Widening the dataset to more states, districts, and crops — plus multiple years of historical data — would make the findings more representative and help confirm real trends
- Water, fertilizer, and pesticide usage were measured here — the next step would be figuring out how to actually optimize them, not just track them
- Including market price data would give a more accurate picture of real profitability, since prices and costs shift over time
- Exploring machine learning models could help forecast future yield trends, instead of only explaining what already happened

GitHub Link

<https://github.com/ttrivedika30-bit/seasonal-agriculture-performance-analysis.git>

VOIS Course completion certificate -

Course name- Data Visualization





Thank you

